

Investigation of oscillation of a spring pendulum

How is the period of the vertical oscillation of a spring pendulum
related to the mass of the pendulum?

Physics Extended Essay

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1 Introduction

A lot of huge science discoveries are inspired from daily simple activities. Newton discovered the law of gravity from an apple falling down a tree. Photoelectric effects are discovered from light bulbs that light up in different colors.

Pendulum is a simple string that hanging back and forth around a fixed point. A spring is also a common material that is used in our daily life, in pen, in boxes, etc. The spring has a property to stretch long and short which is not suitable to use as a rope to connect things. However, this draws my attention. What will happen if these two things combine together: a pendulum with a spring as its rod. This turns out to be a spring pendulum. The period of simple pendulum with an inelastic rod is clear where the period of such spring pendulum remains a mystery for me. I then decided to set up an experiment to find out the period of such system.

After doing some researches, I realize that there are two types of periods that can be considered in a spring pendulum: horizontally and vertically. Since the main property of a spring is its ability to stretch longer and shorter that makes a vertical changing to the height, I decide to investigate on the period of oscillation of the pendulum in its vertical direction.

1.1 Research Question

How is the period of the vertical oscillation of a spring pendulum related to the mass of the pendulum?

2 Exploration

2.1 Theory Deduction

In order to illustrate the system clear, an ideal model of the system is shown below.

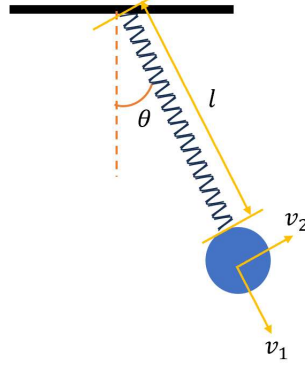


Figure 1: an ideal model of the spring pendulum system

From figure 1, the “zig-zagged” line shows the spring; and the mass is illustrated as a blue circle. The orange straight line is a representation of an imaginary vertical axis. From this figure, it can be clearly shown that there are basically four different variables in this experiment.

l is the length of the spring.

θ is the angle between the spring and the vertical axis.

v_1 is the component of velocity of the mass along the spring.

v_2 is the component of velocity of the mass perpendicular to the spring. The direction of v_2 is perpendicular to the direction of v_1 .

In order to determine these four variables in this system, it is necessary to establish an equation set with four variables and four equations. I figure out that these four variables can all be considered as a function against time. In addition, according to Newton’s Second Law of Motion, the acceleration of an object has a positive relationship to the net force acting upon it; the direction of this acceleration is same to the direction of the net force acting to the object. This can be stated through the equation below,

$$F = m \cdot a \quad (1)$$

Therefore, I decide to combine these four variables through motion mechanism. In this situation, there are two forces acting on the mass (with the neglect of frictions). One is its own gravitational force, acting downward vertically. The other is the tension provided by the stretching of the spring.

Consider the variable l first. The nature of length of the spring is distance from the mass to the fixed hanging point; and the derivative of distance against time is the velocity of the object. In this system, this velocity is the component of velocity of the mass hanging on the spring which is along the direction of the spring. This is also indicated as v_1 that is mentioned above. Hence, we get the first function:

$$\frac{dl}{dt} = v_1 \quad (2)$$

Next, consider the variable θ . The derivative of angle θ against with t is the velocity. Angular velocity is defined as the rate of angular displacement of the mass about a fixed point. This fulfills the condition of angle θ in this situation of the spring pendulum. Moreover, the angular velocity (ω) is derived from:

$$\begin{aligned} v &= \omega \cdot r \\ \omega &= \frac{v}{r} \end{aligned} \quad (3)$$

In this case, v is the component of velocity of the mass perpendicular to the spring, indicated as v_2 since this velocity is responsible for the rotation of the mass. r is the length of the spring l .

Therefore,

$$\frac{d\theta}{dt} = \omega = \frac{v_2}{l} \quad (4)$$

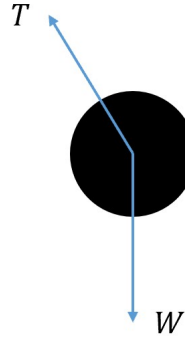


Figure 2: a free-body diagram of the object hanging

Moreover, we also need to consider the acceleration of the mass by analyzing the force on the mass. A free body diagram for the mass hanging is shown in figure 2. Firstly, the gravitational force can be resolve into the direction of the spring, which is equal to:

$$F = m \cdot g \cdot \cos \theta \quad (5)$$

Moreover, due to the Hook's Law^[1], the tension provided by the spring is:

$$\begin{aligned} T &= k \cdot \Delta l \\ T &= k \cdot (l - l_0) \end{aligned} \quad (6)$$

where l_0 is the original length of the spring

Since the direction of tension and component of gravitational force along the spring is opposite, they need to subtract each other. Since the positive direction along the spring is set as the direction away from the fixed point, the derivative of the velocity component along the spring against time can therefore be determined:

By using the Newton's Second Law of Motion^[2],

$$\begin{aligned} F_{net} &= m \cdot a \\ &= W \cdot \cos \theta - T \\ &= W \cdot \cos \theta - k \cdot (l - l_0) \end{aligned}$$

$$a = \frac{W \cdot \cos \theta - k \cdot (l - l_0)}{m} \quad (7)$$

Therefore,

$$\frac{dv_1}{dt} = g \cdot \cos \theta - \frac{k \cdot l}{m} + \frac{k \cdot l_0}{m} \quad (8)$$

At last, consider the derivative of v_2 against t .

$$\begin{aligned} F_{\text{net}} &= m \cdot a \\ &= W \cdot \sin \theta \end{aligned}$$

Similar to the method to derive the acceleration of v_1 ,

$$a = g \cdot \sin \theta \quad (9)$$

In figure 1, the direction of v_2 is oppose to the direction of falling. Thus, the direction of such acceleration should be negative.

$$\frac{dv_2}{dt} = -g \cdot \sin \theta \quad (10)$$

Now consider,

$$\frac{dy(1)}{dt} = g \cdot \cos \theta - \frac{k \cdot l}{m} + \frac{k \cdot l_0}{m} \quad (11)$$

$$\frac{dy(2)}{dt} = -g \cdot \sin \theta \quad (12)$$

$$\frac{dy(3)}{dt} = y(1) \quad (13)$$

$$\frac{dy(4)}{dt} = \frac{y(2)}{y(3)} \quad (14)$$

where,

$$y(1) = v_1$$

$$y(2) = v_2$$

$$y(3) = l$$

$$y(4) = \theta$$

The equations have no analytic solutions but only numerical solutions, so I use MATLAB to solve these equations. The code I written will be shown with detailed explanations in appendix A.

Making the object with 50.13 g mass as an example again, in order to find the theoretical value of oscillation of the spring pendulum along the vertical direction, the vertical displacement of the mass is calculated, which is the product of $\cos \theta$ with l . By importing the data of l and θ from MATLAB into Logger Pro, the relationship of vertical displacement against time can thus be illustrated in figure 3:

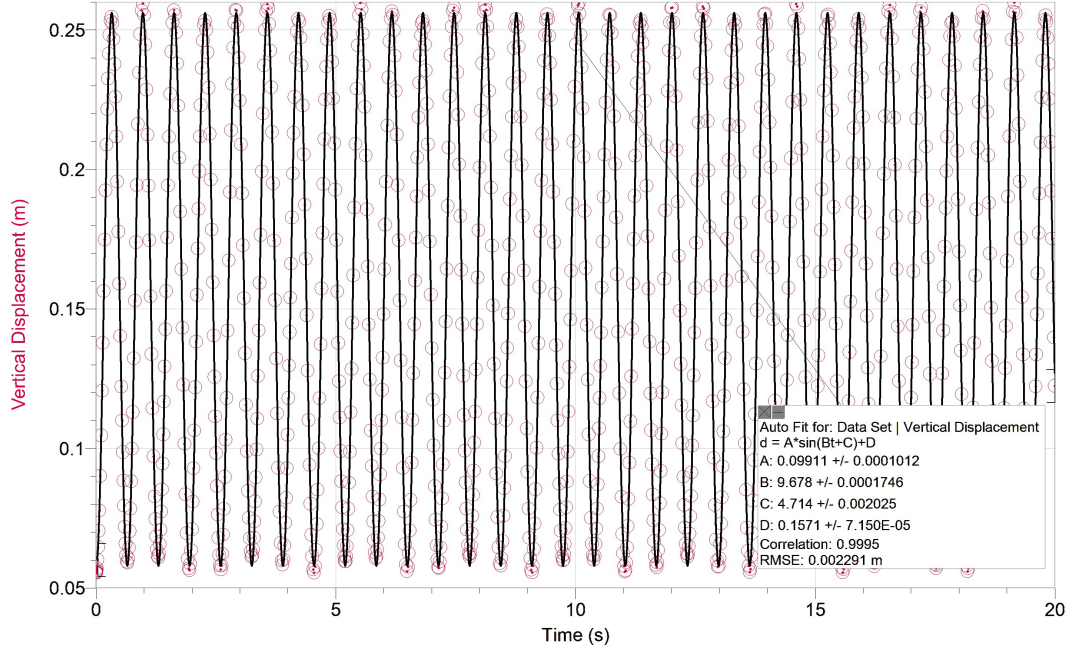


Figure 3: relationship between vertical displacement and time in theory

The sine-fit shows a sine equation of:

$$d = 0.09911 \times \sin(9.678 \times t + 4.714) + 0.1571 \quad (15)$$

Therefore, the value of angular frequency, ω (that is B in the above sine function) can be determined, which are listed in table 1.

Table 1: raw data of the angular frequency of the eight different masses

Trial	Mass/ $10^{-3}kg$	Angular Frequency/ s^{-1}
1	29.59	12.59
2	50.13	9.678
3	69.88	8.197
4	79.73	7.672
5	100.36	6.839
6	149.98	5.595
7	179.55	5.112
8	200.23	4.840

By using mathematics theory, the vertical period for such oscillation can thus be determined:

$$\begin{aligned}
 T &= \frac{2\pi}{\omega} \\
 &= \frac{2\pi}{9.678} \\
 &= 0.64922 \\
 &\approx 0.6492 \text{ s}
 \end{aligned}$$

Similarly, the theoretical value of vertical period of oscillation of other masses hanging on this spring pendulum can be determined through the same way illustrated above. In this experiment, I decide to use *eight different masses* to determine their relationship separately. Before doing the experiment, I am eager to first find out a theoretical relationship between the vertical period of a spring pendulum and the mass hanging on it. After processing the data using method listed above, the theoretical results are listed below in table 2:

Table 2: different vertical period for the eight different masses

Trial	Mass/ $10^{-3}kg$	Vertical Period/ s
1	29.59	0.50
2	50.13	0.649
3	69.88	0.767
4	79.73	0.819
5	100.36	0.919
6	149.98	1.123
7	179.55	1.229
8	200.23	1.298

A graph is then used to show the relationship above geographically:

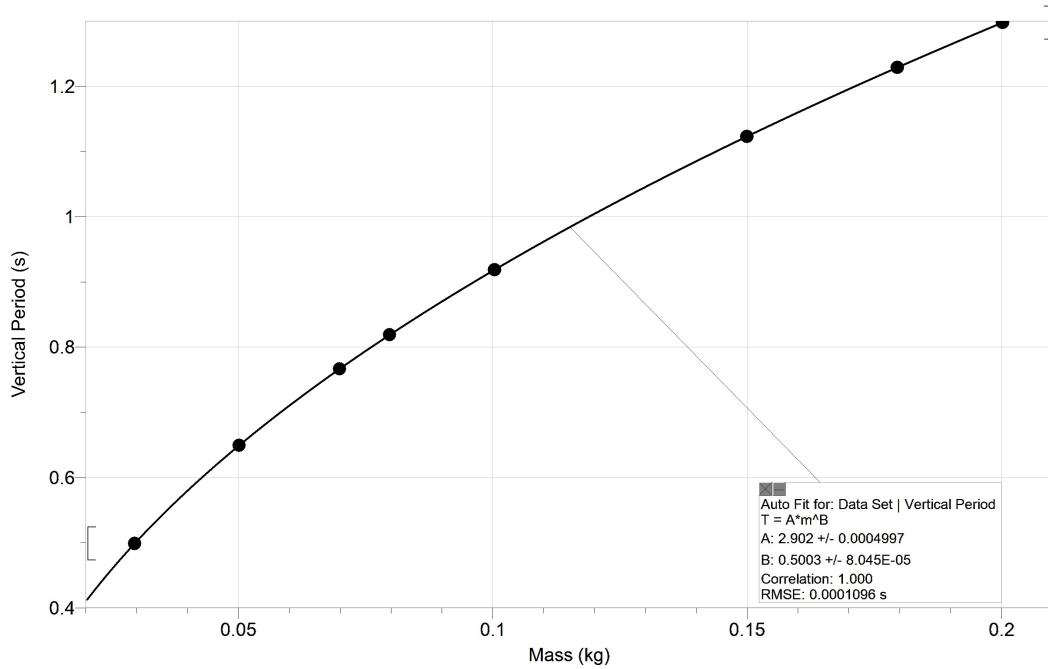


Figure 4: a curve fit for the data set above

Therefore, from figure 4, the theoretical relationship between the vertical period of the oscillation of a spring pendulum and the mass hanging on it fulfills a power function with general equation $y = A \times x^B$. In this case,

$$T = 2.902 \times m^{0.5003} \quad (16)$$

To be more specific, the order of m is 0.5003 which can be approximately considered as 0.5. This shows that in theory, the vertical period of the oscillation, T , has a positive relationship to the square root of the mass hanging on it, $\sqrt{m}$.

$$T \propto \sqrt{m}$$

Thus, my hypothesis is: the period of the oscillation of the spring pendulum in the vertical direction is almost proportional to the square root of the mass hanging to the spring. Next, the ways to determine the value of variables will be shown in detailed through the following paragraphs.

3 Variables

3.1 Independent Variables

The independent variable is the mass of the object that is hanged to the end of the spring. This is measured in kilograms by using an electric balance. During each trial of the experiment, I changed the mass of the object by choosing different weights. The weights are varied from 0.02959 kg to 0.20023 kg.

3.2 Dependent Variables

The dependent variable is the period of oscillation of the spring pendulum in the vertical direction. This is measured through a recording camera. By recording oscillation of the pendulum in 20 s, the video can be loaded into Logger Pro for further analysis. The Logger Pro can separate the horizontal and vertical displacement of the spring pendulum. By fitting the vertical displacement-time graph with sine function, the vertical period of the spring pendulum can thus be determined. The detailed way to do this will be shown in the part of data analysis.

3.3 Controlled Variables

The controlled variables of the experiment are the length of the spring at rest with no mass hanging, the elastic coefficient of the spring and the initial angle where the spring is released.

For the rest length of the spring, it is measured using a ruler. During the whole experiment, there is only one spring that is used. The unit of the length is meter. The value of it measured is 0.064 m.

For the elastic coefficient of the spring, it is measured by using Hook's Law, $F = k \cdot \Delta l$. By placing the spring vertically with a mass hanging at the end, the change of its length can be measured and the force (F) is equal to the weight of the object. Therefore, the elastic coefficient (k) of the spring can be measured. The unit of it is Nm^{-1} . The value measured is $4.798 Nm^{-1}$.

For the initial angle, it is measured in radians. It is measured by using a set square and the value is fixed at $\frac{\pi}{6}$ rad. The detailed ways to control this variable will be shown in the following parts.

4 Methodology

4.1 Apparatus

The apparatus of the whole experiment is listed below,

1. a protractor
2. a spring with an elastic coefficient of 4.798 Nm^{-1}
3. several weights (range from: 5 g, 20 g, 50 g, 200 g)
4. a ruler (range: 1 to 100 cm; minimum division value: 1 mm)
5. an iron-support
6. a wooden stick
7. a camera (a recording device)

4.2 Experimental Set-Up

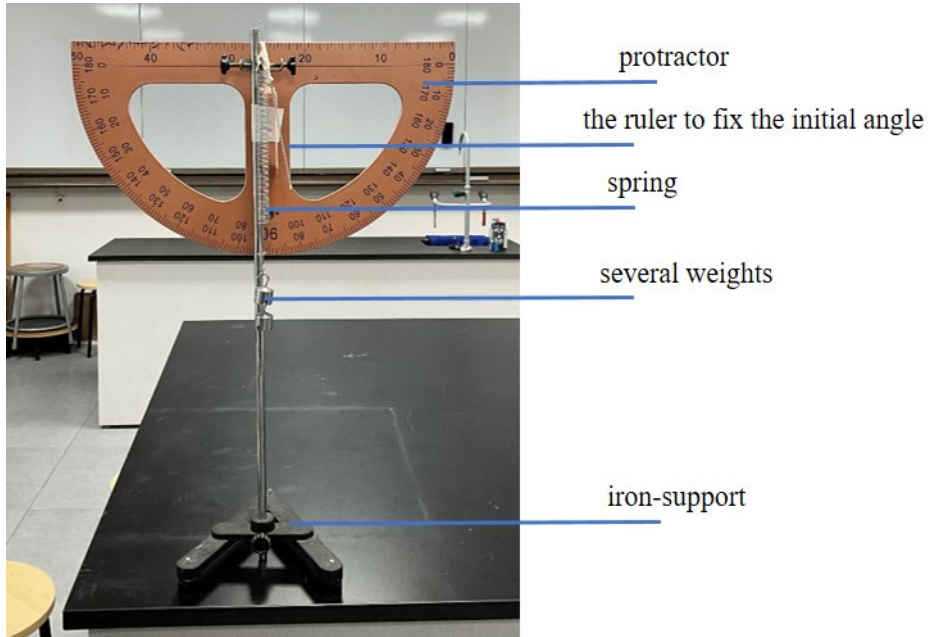


Figure 5: the set-up diagram for the experiment, not to scale

In order to stimulate the environment as ideally as possible, the set up for the apparatus is important. The detailed ways of doing so are listed below. The iron-support is used to hang the spring. Moreover, since the height of an iron-support might not fulfill the stretching of the spring, the spring needs to be fixed onto an extended wooden stick. The wooden stick is the one that is fixed to the iron-support. This stick can reach out of the edge of the table therefore make the space larger for oscillation. In my experiment, I use a tape to tie the spring to the wooden stick, therefore, fixing the spring. In order to control the initial angle, a ruler is stuck to the wooden stick. The angle of the wooden stick to the vertical axis is measured before it is assembled to it. I take the angle at $\frac{\pi}{6}$. Hence,

during each trial of the experiment, the spring can be pushed towards its original length and released at an angle of $\frac{\pi}{6}$ within control. The control variables are all controlled.

4.3 Procedure

Before the beginning of oscillation, a camera is put in front of the iron-support to provide a whole front view of the motion. The weights are hanged to the spring in their equilibrium. After starting the recording, release the weights at $\frac{\pi}{6}$ to the vertical direction with the spring at its original length. Stop recording after the oscillation has lasted for at least 20 seconds. After the system reach equilibrium, repeat these processes for 3 times. Change the mass of the weights for 7 other times and repeat the processes above.

5 Analysis

5.1 Raw Data Collection

In this part, the mass of a 200.23 g weigh is used as an example to show the process of collection of raw data for the experiment. As mentioned in the procedures above, the motion of the spring is recorded through a camera. Therefore, the raw data of such motion is loaded into Logger Pro. By using the function of the video analysis of the software, the varies in the vertical motion of the weights can thus be illustrated.

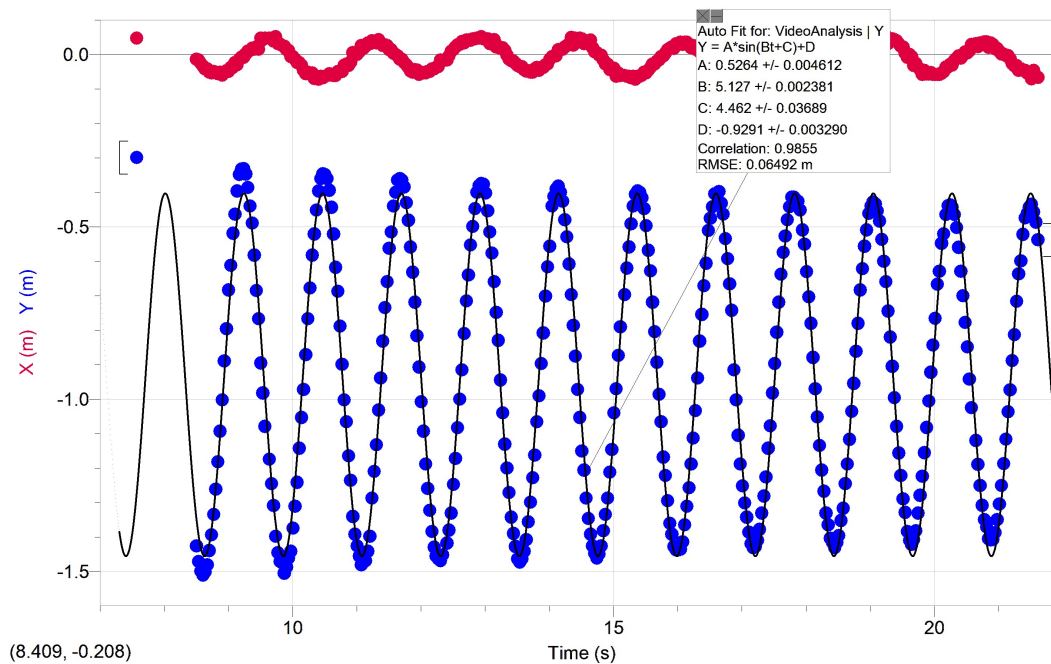


Figure 6: graph shows the motion of the weight at both x and y axis

In figure 6, the red points show the motion along the x-axis, the horizontal axis; while the blue points show the motion along the y-axis, the vertical axis. As mentioned in the research question, what I am interested in is the motion of the spring pendulum along the vertical axis. Hence, the red points can be ignored.

By using the sine function of curve fitting in Logger Pro, a general function of the data collected can be shown. The formula is listed below:

$$y = 0.5264 \times \sin(5.127 \times t + 4.462) - 0.9291 \quad (17)$$

In this equation, the period of this sine function can be obtained by using $T = \frac{2\pi}{\omega}$, where ω is equal to B listed in Figure 6. Therefore, the period of the vertical motion of the spring pendulum with a mass 200.23 g hanging at the end is:

$$\begin{aligned} T &= \frac{2\pi}{\omega} \\ &= \frac{2\pi}{5.127} \\ &= 1.2255 \\ &\approx 1.226 \text{ s} \end{aligned}$$

For each trial of the experiment, I use a same mass that hang on the end of the same spring pendulum and measure the motion for three times repeatedly. Therefore, there should be totally 24 data recording for the whole experiment. By using the method listed above to deal with the raw data, a table is made to illustrated such results:

Table 3: raw data for the whole experiment, containing 24 data in total

Mass/ $10^{-3}kg$	Period 1/s	Period 2/s	Period 3/s	Uncertainty/s
29.59	0.497	0.487	0.502	0.001
50.13	0.630	0.641	0.644	
69.88	0.774	0.787	0.781	
79.73	0.831	0.835	0.835	
100.36	0.939	0.910	0.927	
149.98	1.111	1.094	1.115	
179.55	1.163	1.164	1.161	
200.23	1.226	1.229	1.237	

5.2 Data Analysis

As shown in table 3, the raw data for the experiment has been obtained. It can be shown that each trial of the experiment has three different recordings. This is used in order to reduce the random error of the experiment. Therefore, these three data can be used to calculate an average value which can be a number to illustrate the result for that mass hanging on the spring pendulum. The way to determine such value is shown below.

Making the three different vertical periods for mass 50.13 g as an example:

$$\begin{aligned} \bar{T} &= \frac{T_1 + T_2 + T_3}{3} \\ &= \frac{0.630 + 0.641 + 0.644}{3} \\ &= 0.6383 \\ &\approx 0.638 \text{ s} \end{aligned}$$

In this case, three different trails for a same mass can lead to a reduction in the uncertainty of the experiment. Hence,

$$\begin{aligned}\Delta\bar{T} &= \frac{T_{\max} - T_{\min}}{2} \\ &= \frac{0.644 - 0.630}{2} \\ &= 0.007 \text{ s}\end{aligned}$$

Hence,

$$\bar{T} = (0.638 \pm 0.007) \text{ s} \quad (18)$$

By using the method listed above, all the other seven data can also be processed. Table 4 is created to show the final results.

Table 4: processed data for the vertical period for the eight different masses

Mass/ $10^{-3}kg$	Average Period/ s	Uncertainty of Average Period/ s
29.59	0.495	0.008
50.13	0.638	0.007
69.88	0.781	0.007
79.73	0.834	0.002
100.36	0.93	0.01
149.98	1.11	0.01
179.55	1.163	0.002
200.23	1.231	0.006

As shown in the theory deduction part, the period T has a positive relationship to the square root of the mass, $\sqrt{m}$. Hence, the data listed in table 4 have to engage in further procession in order to linearize the data. Consider the mass of the object 50.13 g as an example again:

Transfer the unit of mass into SI unit first,

$$50.13 \text{ g} = 0.05013 \text{ kg}$$

$$\begin{aligned}\therefore \sqrt{m} &= \sqrt{0.05013} \\ &= \sqrt{0.223897} \\ &\approx 0.22390 \text{ kg}^{\frac{1}{2}}\end{aligned}$$

The uncertainty should be changed as well,

$$\begin{aligned}\Delta m &= 0.00001 \\ \therefore \Delta\sqrt{m} &= \frac{1}{2} \times \frac{\Delta m}{m} \times \sqrt{m} \\ \Delta\sqrt{m} &= \frac{1}{2} \times \frac{0.00001}{0.05013} \times 0.22390 \\ &= 0.000022 \\ &\approx 0.00002 \text{ kg}^{\frac{1}{2}}\end{aligned}$$

Hence,

$$\begin{aligned}\sqrt{m} &= 0.23390 \pm 0.00002 \\ &= (233.90 \pm 0.02) \times 10^{-3} \text{ kg}^{\frac{1}{2}}\end{aligned}\quad (19)$$

Table 5 is thus created to show the further procession of the data in table 4 by using the method illustrated above.

Table 5: processed data for the vertical period for the eight different square root of masses

Square Root of Mass/ $10^{-3} \text{ kg}^{\frac{1}{2}}$	Uncertainty of Square Root of Mass/ $10^{-3} \text{ kg}^{\frac{1}{2}}$	Average Period/s	Uncertainty of Average Period/s
172.02	0.03	0.495	0.008
233.90	0.02	0.638	0.007
264.35	0.02	0.781	0.007
282.37	0.02	0.834	0.002
316.80	0.02	0.93	0.01
387.27	0.01	1.11	0.01
423.73	0.01	1.163	0.002
447.47	0.01	1.231	0.006

5.3 Processed Data Collection and Presentation

In order to verify the theoretical relationship between the vertical period of a spring pendulum and the mass hanging on it that has been illustrated in theory deduction part, a graph can be plotted to show such relationship by using the experimental data. I use the Logger Pro again to achieve this aim, the figure 7 below shows the graph and the function.

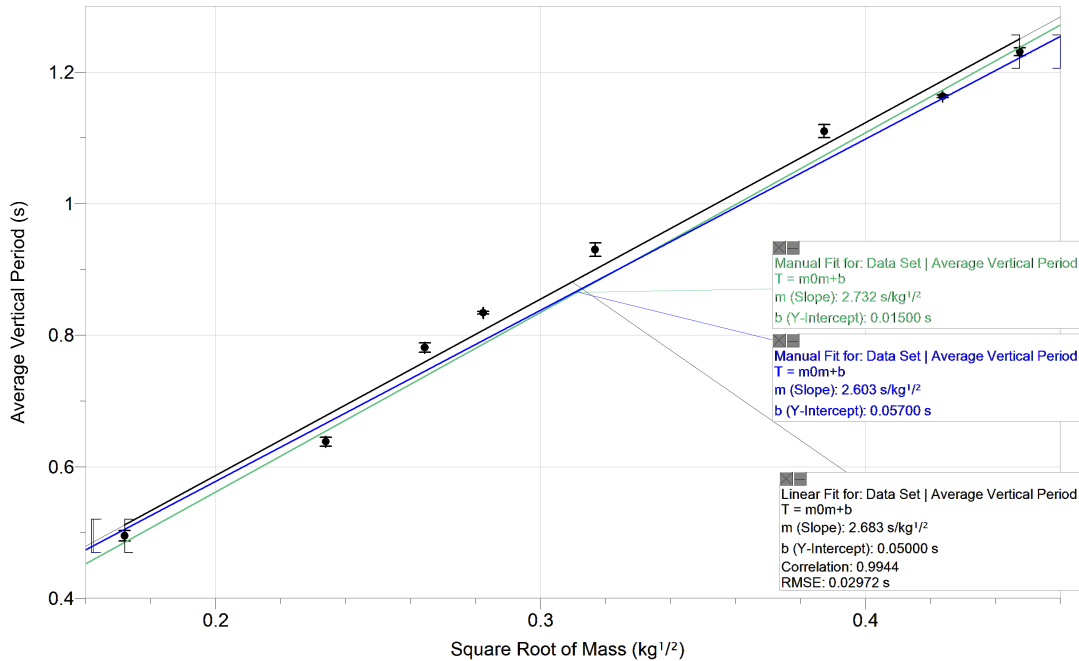


Figure 7: a linearized relationship for the vertical period and square root of mass experimentally

A linear function is also being fit for this data set.

$$T = 2.683 \times \sqrt{m} + 0.05000 \quad (20)$$

In this function,

T is the vertical period of the spring pendulum.

m is the mass hanging on the spring pendulum.

From figure 7, the independent and dependent variables form a linear relationship between each other. This match the hypothesis I made at the beginning of the essay, in the part of the theory deduction. The vertical period of the oscillation is positively related to the square root of the mass hanging at the end of the spring. The best fit line in black in figure 7 has a gradient of $2.683 \text{ s kg}^{-\frac{1}{2}}$ and a y-intercept at 0.05000 s which is not being considered in the theoretical function.

In figure 7, there are also three lines in different colors. The blue and green ones show the maximum gradient line and the minimum gradient line. The green shows the maximum and the blue shows the minimum. The black line therefore shows the best fit line for the experimental data. The maximum gradient is $2.732 \text{ s kg}^{-\frac{1}{2}}$ and the minimum gradient is $2.603 \text{ s kg}^{-\frac{1}{2}}$. Therefore, the percentage uncertainty for the gradient can be calculated:

$$\begin{aligned} \text{Percentage Uncertainty} &= \frac{\frac{m_{\max} - m_{\min}}{2}}{m_{\text{best}}} \times 100\% \\ &= \frac{\frac{2.732 - 2.603}{2}}{2.683} \times 100\% \\ &= 2.404\% \\ &\approx 2.4\% \end{aligned} \quad (21)$$

6 Conclusion

As being mentioned in the Introduction part, the research question for this investigation is “*how is the period of the vertical oscillation of a spring pendulum related to the mass of the pendulum?*” After using physics theories, especially the Newton’s Law of Motion and the Hook’s Law, I set up a differential equation group consists of all the variables in the experiment. By using MATLAB to solve the function group, I figure out a theoretical relationship between the vertical period and the mass hanging on the spring. It shows a positive linear relationship between the vertical period and the square root of the mass hanging. In other words, $T = 2.902 \times \sqrt{m}$. After conducting the experiment, I also get my experimental results, which shows $T = 2.683 \times \sqrt{m} + 0.05000$, after doing data analysis. The characteristics of the graph of the experimental result is also a linear and positive relationship between the independent variable and the dependent variable. Hence, the experimental value match with the hypothesis in the theoretical value.

From the Data Analysis part, the gradient I get from the experiment is 2.683 while it is 2.902 in theory. Thus, a percentage error between the theoretical gradient and the experimental value can thus be determined:

$$\begin{aligned} \text{Percentage Error} &= \frac{2.902 - 2.683}{2.902} \times 100\% \\ &= 7.5465\% \\ &\approx 7.6\% \end{aligned} \quad (22)$$

Therefore, the percentage discrepancy is determined. By comparing it to the percentage uncertainty calculated in the Analysis part, a comparison between the systematic and random error can be determined. The percentage uncertainty is 2.4% which is smaller than the value of percentage discrepancy. Hence, the random error is not as significant as the systematic error.

7 Evaluation

The errors exert that lead to the differences between the theoretical function and the experimental function can be divided into mainly two parts: the systematic errors and the random errors.

7.1 Systematic Errors

By comparing equation 16, the theoretical equation, to equation 20, the experimental equation, there is an additional y-intercept for the experimental value and a smaller gradient.

The first reason that might cause this error is due to the large masses hanging especially during the last 2 trials of the experiment. When the mass increases, the gravitational force exerts on it will increase as well which increases the force experienced by the spring. However, due to the speed of falling which is especially fast at the last 2 trials, the force experienced by the end of the spring cannot evenly distributed to the whole spring. Therefore, the half bottom of the spring will stretch first to provide tension. In other words, the distance between each coil of spring is longer for the half bottom than that for the upper half. Hence, the upper half the spring does not provide the tension which it supposed to do in theory. When I measure the elastic coefficient of the spring at the beginning, I hang the weights to the spring when they are both at rest; in that case, the coil is evenly stretched. During the experiment, the upper half of the spring does not experience the force and thus the spring has lost some of its original length. Hence, the elastic coefficient for the spring is increased in this case due to the Hook's Law, $T = k \cdot \Delta x$. By substituting a larger elastic coefficient to the original function group, a shorter period can be determined.

For the ways to eliminate this problem, a spring that has a larger elastic coefficient can be used. When the mass increases, the tension provided will also increase. Therefore, a larger tension will be used to overcome the large gravitational force of the object. This will decrease the speed of falling and provide the spring enough time to stretch more evenly than it does in previous experiments. Therefore, the upper half of the spring can stretch as well to make the elastic coefficient more constant during the experiment.

The second cause for the systematic error might result from the mass of the spring itself. When measuring the mass, I only consider the mass of the weights. However, when oscillating, the mass of the spring itself is also involved. Therefore, the mass during the oscillation is larger than the mass I measured previously. The mass of the oscillation is been underestimated. Hence, when substituting a larger mass to the original differential function group, a shorter period can be determined.

For the solution to this problem, the mass of the spring itself should be considered when doing the measurement. However, due to the first systematic error, a larger mass will lead to a larger elastic coefficient when oscillating. Hence, when during the experiment, the elastic coefficient of the spring can be increased as well in relative to the increase

of the mass; since during the experiment, the only thing I care about is the period of oscillation. Therefore, such changes will have no effect to the result of my experiment.

The last possible reason for the systematic error might result from the angle when I released the mass. At the end of the spring I used, there is a hook which extends from the end of the spring. The hook has its length as well and is not fixed at the end of the spring. Therefore, when releasing, the hook will not follow the direction of the spring. This situation is especially obvious as the mass increases. The hook will skew downward due to the large gravitational force of the mass hanging. Hence, the actual angle when releasing will decrease and become smaller than the angle in theory. The angle is been underestimated. By substituting a smaller one into the original function, a shorter period can be determined which fulfill the condition results from the systematic error.

This error can also be solved by just remove the hook and hang the object directly at the end of the spring. This can remove the effect of the hook and make the experiment more accurate.

7.2 Random Errors

From figure 7, not all the points on the graph lie on the best-fit line. Therefore, there must be some random errors that reduce the precision of the experiment.

The first random error might result from my action during the video analysis in Logger Pro. During the analysis, I mark all the position of the mass by hand. This will lead to uncertainty which is because it is hard and impossible to notice whether I click on the same place every time during the analysis. Therefore, the point I am analyzing varies from one frame to another. This is random and cause by human analysis. This can thus be considered as a random error.

The way to eliminate this effect is to mark a mark on the weights I used during each different trials. Therefore, the whole mass can be converted onto a single point mass. When clicking in Logger Pro, I can just click on the mark every time, thus make sure that every one of my click is on the same place. This can eliminate this random error significantly.

Another random error might come from the angle when I released the spring. Even though there is a ruler that mark the correct angle to release each time of the experiment, it is impossible to make sure the accuracy of the angle I released. Since I do it manually, there must be some vibrations around the angle $\frac{\pi}{6}$ which is the angle that the spring is released in theory. A very small shift of my hand can lead to a huge shift of the value of the angle. Moreover, it is still hard to ensure the initial speed of the mass; since when releasing my hand might provide the system some external forces. Hence, the initial speed of the system might not be 0 which it supposed to be in theory.

For the ways to solve these errors, mechanical operation can be used instead. An electromagnet can be placed on the right side of the weight that can attract the weight when it is being charged. Thus, at the beginning of the experiment, the electromagnet can be discharged to release the spring automatically. This will make the initial angle same as it is in theory during each trial of the experiment. Moreover, the initial speed can also be controlled and maintain at 0 which fulfills the condition in theory.

References

- [1] Wikipedia Contributors (2019). Hooke's law. [online] Wikipedia. Available at: https://en.wikipedia.org/wiki/Hooke%27s_law.
- [2] Hall, N. (2023). Newton's Laws of Motion. [online] Glenn Research Center. Available at: <https://www1.grc.nasa.gov/beginners-guide-to-aeronautics/newtons-laws-of-motion/>.

Appendix A Code written in MATLAB with explanations

The entire code I written in order to solve the differential equation group (equation 11, equation 12, equation 13, and equation 14) are listed in figure 8. The blue part shows the independent variable of the experiment, the mass of the object hanging on the spring. In the actual code written, $[m]$ will be replaced with the mass of the object measured in kilogram. In other word, in the case of the 50.13 g object, $[m]$ will be replaced with 0.05013. The yellow part shows the constant of the experiment. 9.81 shows the value of the gravitational constant. 4.798 shows the elastic coefficient of the spring. 0.064 shows the length of the spring.

```
function dy=df[001](x,y)
    dy=zeros(4,1);
    dy(1)=9.81*cos(y(4))-4.798*y(3)/[m]+4.798*0.064/[m];
    dy(2)=-9.81*sin(y(4));
    dy(3)=y(1);
    dy(4)=y(2)/y(3);
end
```

Figure 8: code I write in MATLAB about the equation set listed above

In the command window, I use the “ode45” method to solve this differential equation set.

```
fx >> [x,y]=ode45('df[001]', [0,20], [0,0,0.064,pi/6])
```

Figure 9: the code written in the command window

In figure 9, the green part shows the name of the differential equation that should be the same as the one written in figure 8. The yellow part shows the time interval of the experiment. In every trail, I observe the motion and period of the spring pendulum for 20 seconds. Therefore, the time interval should be from 0 second to 20 second. The blue part shows the initial condition of the equation set. Because there are four variables in total, there should be four initial conditions. From left to right, the number shows the initial condition given for v_1 , v_2 , l , and θ . Since the pendulum is at rest at first, v_1 and v_2 should be 0. Since the spring is release at its initial length, l should therefore be its initial length. I choose to release the spring at the angle of $\frac{\pi}{6}$ hence it should be the initial condition for θ .